

Algorithm Aversion in Financial Decision-Making: Effects of Stake Magnitude on Trust

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Abstract

Algorithm aversion — the tendency to distrust automated advice — may intensify when financial decisions carry higher stakes. In a 2 (Advisor Source: AI robo-advisor vs. human financial advisor) $\times$ 2 (Stakes: up to 5% vs. up to 40%) fully within-subjects design, 42 participants imagined choosing between two portfolios and reported their willingness to delegate as well as the influence of the advisor's recommendation on their own decision. High stakes reduced both, but identically for AI and human advisors, and the Source $\times$ Stakes interaction was negligible, with no moderation by experience, task subjectivity, or trust. The findings therefore suggest that high stakes reduce reliance on financial advice generally, not on algorithms specifically.

Introduction

The rise of AI-driven financial tools — robo-advisors, algorithmic portfolio recommenders, and increasingly, general-purpose AI assistants offering financial guidance — has made a long-standing psychological question newly practical: when do people trust machines with their money? This question is especially pressing for younger investors, who are entering financial markets earlier and more independently than previous generations, often through platforms that pair investment tools directly with AI-generated recommendations. Whether this population defers to algorithmic advice, and under what conditions that deference breaks down, has direct implications for how such tools should be designed, disclosed, and regulated. If algorithmic reluctance is strongest precisely when the financial stakes are highest — retirement allocations, major restructurings, decisions with long-term, hard-to-reverse consequences — then the moments when algorithmic tools could do the most good for consumers may be exactly the moments consumers are least willing to use them.

To understand why source identity matters during decision making, it is useful to begin with the classical economic assumption it violates. Traditional rational-choice theory imagines a decision-maker with complete information, a stable set of preferences, and the computational capacity to evaluate every alternative before choosing the one that maximizes expected utility. Simon (1955) challenged this assumption directly, arguing that real decision-makers are bounded by limited knowledge, limited cognitive processing capacity, and limited time, and therefore cannot optimize in the way classical theory assumes. In place of optimization, Simon proposed satisficing: rather than searching exhaustively for the single best option, individuals simply search for one that clears an aspiration threshold. This reframing is important because it establishes that heuristics and shortcuts are not deviations from rationality that a smarter or more motivated person could avoid — they are a structural feature of how any cognitively limited agent must operate. Source identity (for example, whether advice comes from a human or a machine) is exactly this kind of shortcut: it allows a decision-maker to sidestep

the effortful work of fully evaluating the quality of advice by instead asking a simpler, faster question about who or what produced it.

Kahneman (2003) extended this foundation into a more detailed architecture of judgment. He distinguished System 1, which is fast, automatic, associative, and effortless, from System 2, which is slow, deliberate, and effortful. He then argued that most everyday judgments are governed by System 1, with System 2 intervening only intermittently to monitor or correct it. Central to this account is the idea of accessibility: some thoughts, cues, and associations come to mind far more readily than others, and System 1 judgments are shaped by whatever features of the situation are most accessible, while less accessible features are largely ignored — even when a full consideration of the evidence would treat them as relevant. Kahneman further showed that this accessibility-driven processing is what underlies framing effects — the same decision, described in different terms, can produce different choices, because different framings make different features of the problem accessible. Accordingly, whether advice is attributed to a robo-advisor or a human advisor is itself a framing manipulation: it changes what associations become accessible when a person is asked whether to follow the advice, independent of the content of the advice itself.

Kahneman and Tversky's (1979) prospect theory supplies the second theoretical pillar. Where expected utility theory assumes decision-makers evaluate outcomes in terms of final wealth states, prospect theory shows that people instead evaluate outcomes as gains and losses relative to a reference point, and that the resulting value function is steeper for losses than for equivalent gains — a phenomenon known as loss aversion. Applied to the present context, prospect theory generates a specific prediction: because losses loom larger than equivalent gains, a decision framed around a potential large loss should recruit a qualitatively different, more emotionally charged evaluation process than the same decision framed around a smaller one. If algorithmic and human advisors are treated differently by that more emotional process, the gap between them should widen precisely as the stakes, and therefore the potential losses, grow.

Loewenstein et al. (2001) make this emotional channel explicit by proposing a “risk-as-feelings” hypothesis. They argue that most theories of risky choice remain cognitive and consequentialist: they assume people assess the probability and desirability of outcomes and integrate this information through some expectation-based calculation. The risk-as-feelings hypothesis proposes an alternative pathway, showing that emotional reactions to risky situations — anticipatory emotions experienced in the moment of decision, distinct from the emotions one expects to feel afterward — often diverge from these cognitive assessments, and that when the two diverge, behavior tends to follow feeling rather than the calculation. This gives a concrete psychological mechanism for why stakes might interact with advisor source: if an algorithm is perceived as calculated and emotionally neutral while a human advisor is perceived as capable of reassurance, shared responsibility, or empathy, then rising stakes — which by Loewenstein et al.’s account intensify the anticipatory emotional response — could differentially erode trust in the source perceived as less equipped to address that emotional experience.

Within this theoretical scaffolding, the specific empirical record on algorithmic advice is contested rather than settled, and understanding that contest is essential to situating the present study. Dietvorst et al. (2015) coined the term algorithm aversion to describe a striking asymmetry: across five forecasting experiments, participants who watched an algorithm err disproportionately abandoned it in favor of human judgment, even when the algorithm continued to outperform the human forecaster overall. Participants who had no prior experience with either forecaster, by contrast, preferred the algorithm over the human — meaning the aversion was triggered specifically by watching the algorithm fail, not by a prior belief that algorithms are untrustworthy.

Complicating this picture, Logg et al. (2019) documented the opposite default pattern, which they termed algorithm appreciation. Across six experiments using an advice-taking paradigm, where the content of the advice was held identical and only its labeled source (“algorithm” versus “person”) varied, laypeople

consistently weighted advice more heavily when they believed it came from an algorithm. This appreciation persisted across several task types but waned under two conditions: when participants had to choose between an algorithm's estimate and their own judgment rather than an external human advisor's, and, more strikingly, among people with genuine forecasting expertise. Experienced professionals who made forecasts regularly in their own work relied on algorithmic advice less than laypeople did — a pattern that paradoxically hurt the accuracy of their forecasts. Together, Dietvorst et al. and Logg et al. establish that neither “people avoid algorithms” nor “people prefer algorithms” is a stable, universal description; the direction of the effect depends on features of the situation.

Castelo et al. (2019) proposed a framework that reconciles these findings. Across four laboratory studies and two large field experiments, they showed that algorithms are trusted and relied on less specifically for tasks perceived as subjective in nature, and more for tasks perceived as objective and quantifiable. Critically, they also demonstrated that this perceived objectivity is malleable: interventions that increase a task's perceived objectivity increase trust in and use of algorithms for that same task, because people mistakenly believe algorithms lack the capacity to handle subjective, affectively loaded judgments. This task-dependent account provides a shared explanatory framework for algorithm aversion and algorithm appreciation: which pattern emerges depends on where a given task sits on the subjective–objective spectrum, as perceived by the person making the decision.

Personal finance sits in an unusually unresolved position on this spectrum. It is structurally quantitative — expected returns, historical yields, volatility, and probability of loss are all calculable in principle — which by Castelo et al.'s framework should favor algorithmic trust. At the same time, personal finance is simultaneously entangled with long-term security, identity, and emotionally loaded concerns about one's own competence and future, which may instead favor a human advisor perceived as capable of empathy and shared accountability. The empirical record on financial delegation reflects this tension rather than resolving it. Filiz et

al. (2022), testing university students across four portfolio-construction tasks against a demonstrably optimal robo-advisor, found that participants entrusted their decisions to the algorithm in only around 40% of cases despite its clear superiority. Germann and Merkle (2023), by contrast, ran an experiment in which young-adult participants tied their incentives to either a human fund manager or an investment algorithm, and found no sign of algorithm aversion at all: participants cared about realized returns but showed no systematic preference for which source produced them, and were not quicker to lose confidence in the algorithm after watching it err, in direct tension with Dietvorst et al.'s (2015) central finding. Prah and Van Swol (2017) add a further nuance: in a domain-general forecasting task, they failed to replicate the finding that human advice is utilized more than automated advice in initial uptake, but found that after receiving bad advice, reliance on the algorithm fell far more sharply than reliance on the human advisor did — the same asymmetry Dietvorst et al. (2015) reported. Algorithm aversion, they suggest, depends less on first impressions than on how differently people recover from a source's mistakes. Together, these three studies suggest that whether algorithm aversion appears in financial delegation tasks depends heavily on design features that are still poorly understood — including, plausibly, how much is at stake in the decision being delegated.

Filiz et al. (2023) supply the most direct evidence connecting stakes specifically to algorithm aversion, describing what they call the “tragedy of algorithm aversion”: using a framing experiment across six decision scenarios with identical probabilities that varied in narrative gravity, they found that the more serious the consequences of a decision, the more frequently participants exhibited algorithm aversion, even though the algorithm's superiority remained constant throughout. This result is important, but its design leaves a key gap unaddressed: gravity was varied by comparing across different scenario narratives rather than by manipulating stakes magnitude within a single, held-constant decision domain. This leaves open the possibility that what looks like a stakes effect is partly an artifact of incidental differences between the scenarios being compared, rather than a pure effect of stakes magnitude itself.

There is also reason to think this stakes-driven aversion, if real, need not be permanent or fixed. Dietvorst et al. (2018) showed across three incentivized forecasting studies that participants were considerably more willing to use an imperfect algorithm when they were permitted to modify its output — and this preference for modifiability persisted even when the permitted modifications were restricted to a small fraction of the forecast. Their interpretation is that algorithm aversion is driven substantially by a felt loss of control and autonomy, not by algorithmic performance itself, since even token control was sufficient to increase adoption and improve outcomes. This raises the possibility that any stakes-driven widening of the AI–human reliance gap in the present study reflects not simply “algorithms are trusted less when it matters more” but something more specific: a loss of perceived control that becomes more aversive as the consequences of losing that control grow larger.

Underlying all of these advisor-specific effects is a more basic question of trust, and Mayer et al. (1995) provide the relevant grounding here. They define trust as the willingness to be vulnerable to another party based on the expectation that the party will act in one’s interest, and they argue that this willingness rests on perceptions of the advisor’s trustworthiness — most centrally its competence (does this advisor know what it’s doing?) and benevolence (does this advisor have my interests at heart?). This motivates decomposing trust into its component parts, since a stakes-driven widening of the AI–human gap could in principle be driven by either, or both, and the two carry different implications for how algorithmic financial tools might be designed to close that gap.

The present study addresses the gap identified above by holding the financial domain constant and manipulating stakes magnitude directly, within-subjects, while independently varying advisor source (AI robo-advisor vs. human financial advisor). We test whether the AI–human reliance gap widens as stakes increase (H1), and whether this pattern is moderated by individual differences in self-perceived investing experience (H2), trust in the advisor (H3), and perceived subjectivity of the investment task (H4). Reliance is

operationalized through two complementary measures: delegation (willingness to hand the decision over entirely) and influence (the degree to which a recommendation shifts a stated choice), allowing us to distinguish whether stakes and source affect the willingness to cede control differently from how much a recommendation is actually weighted. Because prior work is genuinely split — Filiz et al. (2022) and Filiz et al. (2023) suggesting aversion intensifies with stakes and consequence, Germann and Merkle (2023) finding no aversion at all, and Prahll and Van Swol (2017) finding aversion concentrated specifically in post-error recovery rather than baseline uptake — this study is positioned not to confirm a predetermined direction but to adjudicate directly, within one financial domain, whether stakes magnitude on its own is sufficient to produce the differential erosion of algorithmic trust that the “tragedy of algorithm aversion” implies.

Method

Participants

Participants were recruited using convenience sampling from the student population of the Singapore University of Technology and Design (SUTD), via a survey link posted on the university’s Telegram group. No demographic information was collected, and participation was voluntary. Participants were offered a \$1 incentive for completing the survey, for which they were asked to optionally provide their phone number.

Design

The study employed a 2 (Advisor Source: AI robo-advisor vs. human financial advisor) $\times$ 2 (Stakes: low vs. high) fully within-subjects design. All participants completed all four conditions. Stakes were operationalised as the maximum potential gain or loss of the investment decision: up to 5% of the initial investment in the low-stakes condition and up to 40% in the high-stakes condition. Two dependent variables were measured: willingness to delegate the decision to the advisor (delegation) and the influence of the advisor's

recommendation on the participant's choice (influence). Three individual-difference variables were measured as potential moderators: self-rated investment experience, perceived subjectivity of the task, and trust in the advisor. The presentation order of conditions was fixed: participants always encountered the AI-advisor block before the human-advisor block, and within each block, the low-stakes scenario before the high-stakes scenario.

Measures

Six measures were administered in the order listed below: perceived investment experience, perceived subjectivity of the task, anxiety, trust in the advisor, willingness to delegate the decision, and influence of the advisor's recommendation.

Perceived investment experience was measured with a single item asking participants "How experienced do you consider yourself in investing?" Responses were recorded on a 5-point scale ranging from 1 (not at all) to 5 (very much).

Perceived subjectivity of the task was measured with a single item: "Do you think this decision (choosing between two portfolios) requires more subjective evaluation (i.e. intuition and experience) or more objective evaluation (i.e. mathematics and data)?" rated on a 5-point scale from 1 (mostly objective) to 5 (mostly subjective).

Anxiety was measured with a single item: "How anxious does this decision make you feel?" rated on a 5-point scale from 1 (not at all) to 5 (extremely anxious). Anxiety was rated once per stakes level, prior to the introduction of either advisor, and therefore served as a manipulation check for the stakes manipulation.

Trust in the advisor was measured with two items per source, based on the competence–benevolence distinction: competence ("I believe that [robo-advisors / human financial advisors] perform their intended function well") and benevolence ("I believe that [robo-advisors are designed to work in my interest / human financial advisors work in my interest]"). Each item was rated on a 5-point scale from 1 (strongly disagree) to 5 (strongly agree), and the two items were averaged to form a composite trust score for each source.

Willingness to delegate the decision was measured with a single item per scenario: "I would allow the [robo-advisor / human financial advisor] to make the decision on my behalf." rated on a 6-point scale from 1 (strongly disagree) to 6 (strongly agree). The scale deliberately omitted a neutral midpoint to force a directional response.

Influence of the advisor's recommendation on the participant's choice was measured with a single item per scenario: "[Advisor] recommends portfolio [X]. Are you more or less likely to choose portfolio [X]?" rated on a 7-point scale from 1 (much less likely) to 7 (much more likely), where 4 corresponded to no change in likelihood. Each scenario presented two arbitrarily labelled portfolios (A/B, C/D, E/F, G/H); the recommended portfolios (A, D, F, G) were randomly selected and carried no further meaning.

Procedure

The study was administered as an online Google Forms survey (see Appendix A) which participants completed independently on their own personal devices. Participants were first asked to imagine that they held a sizable amount of money in a reputable investment platform and were choosing between two portfolios with the goal of maximising their return after one year. They then completed the two covariate items (experience and perceived subjectivity), followed by the anxiety ratings for the low-stakes and then the high-stakes decisions. Next, participants completed the trust items for the AI robo-advisor and responded to two scenarios for this advisor — one low-stakes and one high-stakes, in that order. The same procedure was then repeated for the human financial advisor. Within each scenario, participants reported both their willingness to delegate and the influence of the recommendation on their choice. Apart from the advisor label and portfolio label, the scenario wording was identical across conditions, and the low- and high-stakes scenarios differed only in the magnitude of the potential gain or loss.

Results

Forty-two participants completed the survey, with 34 accepting the financial incentive. All 42 participants were included in the analyses, with raw data presented in Appendix B.

Self-reported anxiety was higher for the high-stakes decision ($M = 3.83$, $SD = 0.96$) than for the low-stakes decision ($M = 2.21$, $SD = 1.02$), a significant difference in a paired comparison, $t(41) = 11.54$, $p < .001$ (Figure 1). The stakes manipulation was effective.

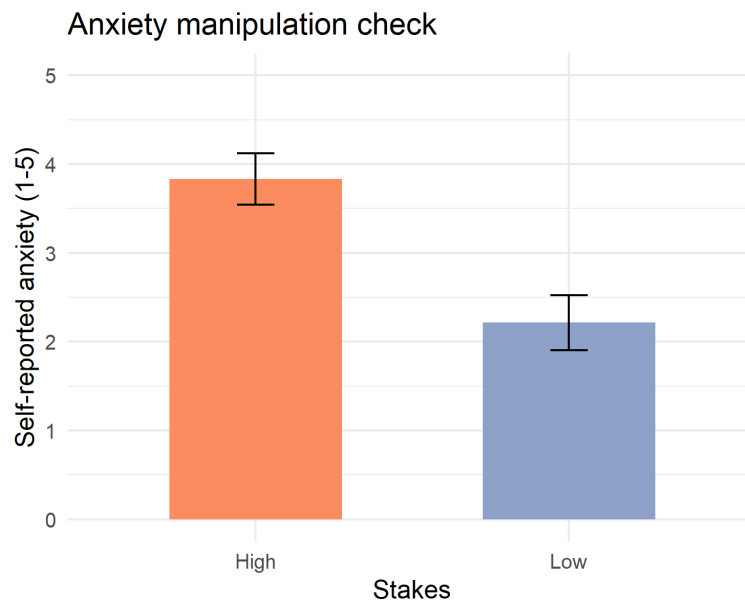


Figure 1. Self-reported anxiety by stakes level. Error bars show 95% confidence intervals.

Cell means and standard deviations are shown in Table 1. Mean scores were lower at high stakes for both sources, and the AI advisor scored slightly lower than the human advisor in every cell (Figures 2 and 3).

Dependent variable	Source	Low stakes	High stakes
Delegation (1–6)	Human	4.05 (1.13)	2.90 (1.32)
Delegation (1–6)	AI	3.76 (1.25)	2.45 (1.25)
Influence (1–7)	Human	5.12 (1.31)	4.33 (1.44)
Influence (1–7)	AI	4.90 (1.01)	3.93 (1.42)

Table 1. Means (SDs) for delegation and influence by advisor source and stakes.

The interaction plots (Figures 2 and 3) display these patterns on the original response scales.

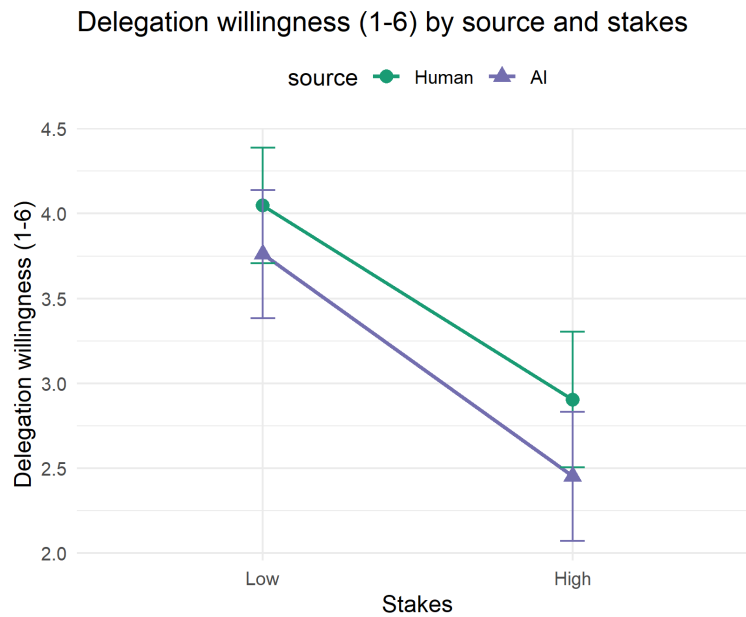


Figure 2. Mean delegation by advisor source and stakes level. Error bars show 95% confidence intervals.

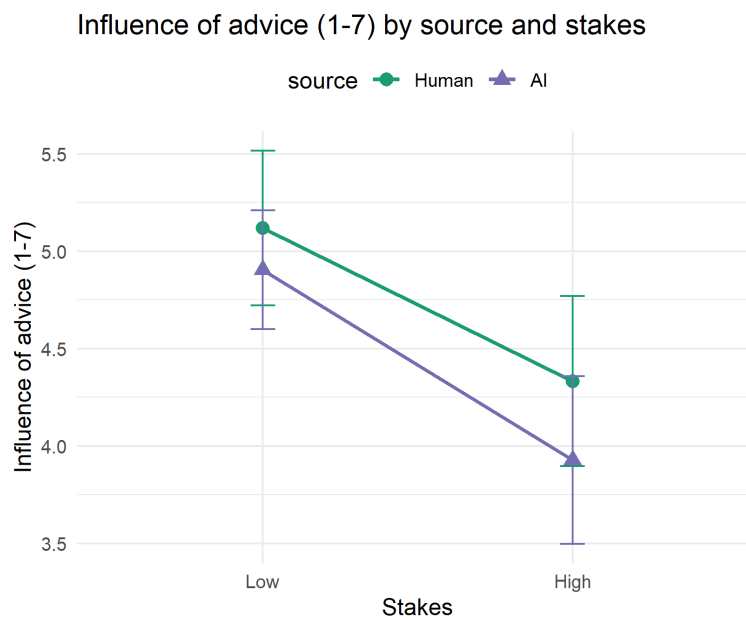


Figure 3. Mean influence by advisor source and stakes level. Error bars show 95% confidence intervals.

The hypotheses were tested with linear mixed-effects models (LMMs) with a random intercept per participant, analysing all 168 observations (42 participants $\times$ 2 sources $\times$ 2 stakes levels). Source was coded

with Human as the reference level, and Stakes with Low as the reference level, so coefficients express the difference between AI and Human and between High and Low stakes. Continuous moderators were grand-mean centred. Fixed effects are reported below for the terms relevant to each hypothesis.

For delegation, the stakes effect was significant and negative, $b = -1.143$, $SE = 0.202$, $p < .001$: delegation was substantially lower at high stakes. The source effect was not significant, $b = -0.286$, $SE = 0.202$, $p = .160$, and neither was the Source $\times$ Stakes interaction, $b = -0.167$, $SE = 0.286$, $p = .561$.

For influence, the same pattern emerged: a significant negative stakes effect, $b = -0.786$, $SE = 0.216$, $p < .001$, and non-significant source, $b = -0.214$, $SE = 0.216$, $p = .324$, and interaction effects, $b = -0.190$, $SE = 0.306$, $p = .535$.

Effect sizes for the H1 models are shown in Figure 4. Partial eta-squared was large for stakes (Delegation: $\eta^2p = .374$; Influence: $\eta^2p = .212$), small for source ($\eta^2p = .051$; .032), and negligible for the interaction ($\eta^2p = .003$; .003). The coefficient plot (Figure 5) summarises all model estimates.

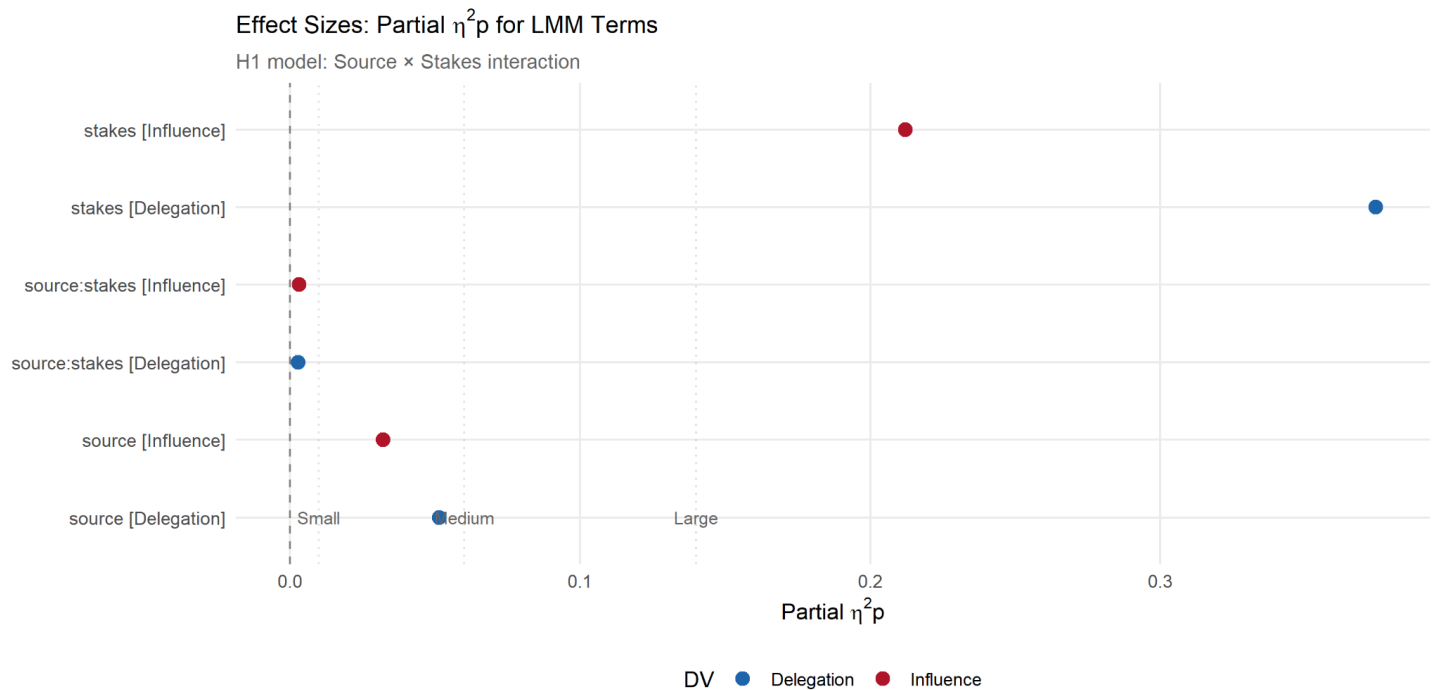


Figure 4. Partial eta-squared effect sizes for the H1 models by dependent variable. Vertical lines mark the conventional small (.01), medium (.06) and large (.14) thresholds.

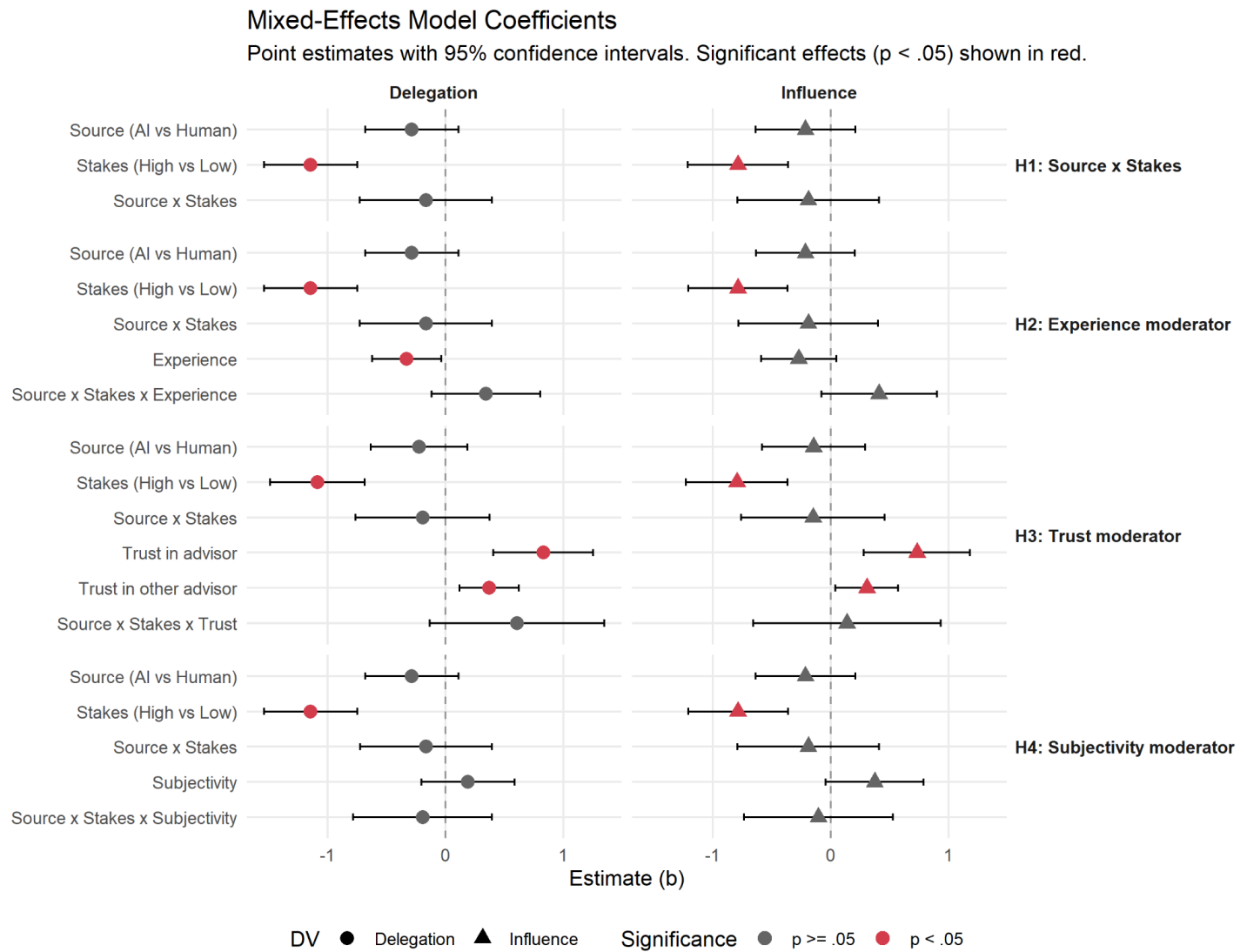


Figure 5. Forest plot of all model coefficients (H1–H4) by hypothesis panel and dependent variable. Red = significant ($p < .05$), grey = non-significant; circles = delegation, triangles = influence; bars = 95% confidence intervals.

More experienced participants delegated less overall, $b = -0.329$, $SE = 0.150$, $p = .030$; the analogous effect on influence was not significant, $b = -0.272$, $SE = 0.162$, $p = .097$. The Experience $\times$ Source $\times$ Stakes interaction was not significant for either DV (Delegation: $b = 0.342$, $SE = 0.235$, $p = .148$; Influence: $b = 0.410$, $SE = 0.250$, $p = .103$).

Trust in the relevant advisor was a strong positive predictor of delegation, $b = 0.829$, $SE = 0.215$, $p < .001$, and of influence, $b = 0.730$, $SE = 0.230$, $p = .002$. Trust in the other advisor also predicted both outcomes

(Delegation: $b = 0.369$, $SE = 0.128$, $p = .005$; Influence: $b = 0.305$, $SE = 0.136$, $p = .027$). The Trust $\times$ Source $\times$ Stakes interaction was marginal on delegation and non-significant on influence (Delegation: $b = 0.606$, $SE = 0.378$, $p = .112$; Influence: $b = 0.138$, $SE = 0.405$, $p = .734$).

Perceived subjectivity (higher scores indicating the task was rated as more subjective) did not significantly predict delegation, $b = 0.191$, $SE = 0.202$, $p = .347$, or influence, $b = 0.372$, $SE = 0.212$, $p = .082$. The Subjectivity $\times$ Source $\times$ Stakes interaction was non-significant for both DVs (Delegation: $b = -0.193$, $SE = 0.300$, $p = .520$; Influence: $b = -0.104$, $SE = 0.321$, $p = .746$).

Discussion

Across both DVs — willingness to delegate and susceptibility to influence — the results were largely null for our main hypotheses. The one consistent exception was that stakes had a strong and consistent main effect on both outcomes.

H1: Source $\times$ Stakes Interaction

Contrary to H1, the Source $\times$ Stakes interaction was not significant for delegation ($b = -0.167$, $p = .561$) nor influence ($b = -0.190$, $p = .535$). AI advisors scored slightly lower than human advisors in all cells, but these differences were small (Delegation: $\eta^2p = .051$; Influence: $\eta^2p = .032$) relative to the large drop in reliance on both advisors at high stakes (Delegation: $\eta^2p = .374$; Influence: $\eta^2p = .212$), and the interaction explained a negligible 0.3% of the variance ($\eta^2p = .003$).

This pattern points to general advice withdrawal under high stakes rather than algorithm aversion specifically: when the potential loss is large, participants refrain from external advisors altogether. This is consistent with Simon's (1955) bounded rationality — individuals under pressure simplify rather than optimize — and with Loewenstein et al.'s (2001) "risk-as-feelings" model, in which anticipatory emotion drives behavior

independently of cognitive assessment. The manipulation check confirmed that high stakes were experienced as more anxiety-provoking, though anxiety was not tested as a mediator, so this link remains suggestive.

This null interaction directly contrasts with Filiz et al. (2023), who found that greater consequences increased algorithm aversion. Two design differences help reconcile this: Filiz et al. varied the gravity of stakes across different scenarios rather than within one held-constant decision domain, and explicitly demonstrated the algorithm as superior (70% vs 60% probability of success) with performance-contingent payment. Their results may therefore partly reflect differences between scenarios rather than stakes alone, an issue our within-subjects design addresses directly. With no comparative performance information and purely hypothetical outcomes, participants in our study had no basis for distrusting the algorithm specifically; which instead aligns with Germann and Merkle (2023), who found no algorithm aversion in delegated investing, and Prahla and Van Swol (2017), who found no baseline discounting of automation without feedback.

H2: Experience as Moderator

The three-way Experience $\times$ Source $\times$ Stakes interaction was not significant for either outcome (Delegation: $p = .148$; Influence: $p = .103$), with coefficients running opposite to the predicted direction. Experience did predict lower delegation overall ($b = -0.329$, $p = .030$). This is consistent with Logg et al.'s (2019) finding that experienced professionals take less advice overall than laypeople. However, in the present study, experience did not significantly interact with advisor source or stakes. H2 was therefore not supported.

H3: Trust as Moderator

The Trust $\times$ Source $\times$ Stakes interaction was also non-significant for both outcomes (Delegation: $p = .112$; Influence: $p = .734$), thus H3 was suggestive but not confirmed. Nonetheless, trust in the advisor was the strongest individual predictor found, strongly predicting delegation ($b = 0.829$, $p < .001$) and influence ($b = 0.730$, $p = .002$), consistent with Mayer et al.'s (1995) integrative model of trust. Notably, trust in the other

advisor also positively predicted both outcomes, suggesting that delegation reflects a general disposition to trust advisors rather than distrust of AI specifically. What this suggests is that if low robo-advisor uptake reflects general distrust of advisors rather than distrust of AI specifically, improving the algorithm's performance may address the wrong target; building general credibility may be more effective. One nuance is worth noting: the human advisor was rated more competent ($M = 3.57$) than benevolent ($M = 3.19$), whereas the robo-advisor's competence ($M = 3.07$) and benevolence ($M = 3.10$) ratings were nearly identical, suggesting that participants questioned whether human advisors act in their interest even while trusting their competence.

H4: Perceived Subjectivity as Moderator

Neither the main effect of perceived subjectivity nor the three-way Subjectivity $\times$ Source $\times$ Stakes interaction was significant (all $p > .08$), so H4 was not supported. This is nonetheless consistent with Castelo et al.'s (2019) broader finding that algorithms are trusted and relied on less for tasks perceived as subjective than for tasks perceived as objective. The investment task was rated largely objective — 27 of 42 participants (64%) rated the investment decision as at least somewhat objective, which produced relatively little variation at the subjective end of the scale, reducing its sensitivity to moderation.

Limitations

Several limitations qualify these conclusions. First, condition order was fixed: the AI block always preceded the human block, and low stakes always preceded high stakes — carryover effects are possible. Second, the convenience sample of SUTD students, drawn from a heavily AI-focused institution, is unlikely to generalise well beyond a young, tech-savvy population. Third, hypothetical decisions are unlikely to carry the genuine emotional consequences of real financial loss that Loewenstein et al. (2001) identify as a key driver of behavior. Finally, with only 42 participants, the study had limited power to detect small-to-medium three-way

interactions, so the H2–H4 nulls should be viewed cautiously; the H1 interaction, with its negligible effect size ($\eta^2p = .003$), is a partial exception.

Future research should randomize the order of advisors and stakes to eliminate carryover bias, recruit a larger, more diverse sample, and add real incentives and validated multi-item scales. In addition, an error-exposure phase, in which participants observe the advisor make a mistake, would test whether algorithm aversion emerges only after an error is observed, as Prahl and Van Swol (2017) and Dietvorst et al. (2015) suggest, rather than as an initial baseline response.

Conclusion

Algorithm aversion — the tendency to distrust automated advice — is a significant barrier to the adoption of AI-driven financial tools, and prior research suggested it may intensify as decision stakes rise. This study tested that prediction directly, holding the financial domain constant while manipulating advisor source (AI robo-advisor vs. human financial advisor) and stakes magnitude within subjects. The findings did not support a stakes-driven form of algorithm aversion: high stakes reduced reliance on advice, but equally for AI and human advisors. Rather than distrust of algorithms specifically, the results point to a general withdrawal from financial advice when the consequences of a decision are high — a conclusion with implications for how AI-driven financial tools should be designed and communicated.

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Appendix A

Pre-survey

1. How experienced do you consider yourself in investing?
☐ not at all ☐ a little ☐ somewhat ☐ fairly ☐ very much

In all of the following scenarios, imagine that you have a sizable amount of money in a reputable investment platform and are choosing between two portfolios with the goal of maximising your return after one year. A portfolio is a collections of financial assets like stocks and bonds. Portfolios differ from one another in terms of diversification and asset allocation.

2. Do you think this decision (choosing between two portfolios) requires more: subjective evaluation (i.e. intuition and experience) or more objective evaluation (i.e. mathematics and data)?
☐ mostly objective ☐ somewhat more objective ☐ equally balanced ☐ somewhat more subjective ☐ mostly subjective

Imagine that you are struggling to choose between two portfolios. Based on your choice, you could gain or lose up to 5% of your initial investment.

3. How anxious does this decision make you feel?
☐ not at all ☐ slightly anxious ☐ moderately anxious ☐ very anxious ☐ extremely anxious

Imagine that you are struggling to choose between two portfolios. Based on your choice, you could gain or lose up to 40% of your initial investment.

4. How anxious does this decision make you feel?
☐ not at all ☐ slightly anxious ☐ moderately anxious ☐ very anxious ☐ extremely anxious

Scenario 1 – Robo-advisor

Suppose that the platform provides you with a free robo-advisor which can analyse your situation using algorithms and give financial advice.

5. I believe that robo-advisors perform their intended function well.
☐ strongly disagree ☐ disagree ☐ neutral ☐ agree ☐ strongly agree

6. I believe that robo-advisors are designed to work in my interest.
☐ strongly disagree ☐ disagree ☐ neutral ☐ agree ☐ strongly agree

Scenario 1.1

Imagine that you are struggling to choose between two portfolios, A or B. Based on your choice, you could gain or lose up to 5% of your initial investment.

7. I would allow the robo-advisor to make the decision on my behalf.
☐ strongly disagree ☐ disagree ☐ slightly disagree ☐ slightly agree ☐ agree ☐ strongly agree

8. The robo-advisor recommends portfolio A. Are you more or less likely to choose portfolio A?
☐ much less likely ☐ somewhat less likely ☐ slightly less likely ☐ neutral ☐ slightly more likely ☐ somewhat more likely ☐ much more likely

Scenario 1.2

Imagine that you are struggling to choose between two portfolios, C or D. Based on your choice, you could gain or lose up to 40% of your initial investment.

9. I would allow the robo-advisor to make the decision on my behalf.
☐ strongly disagree ☐ disagree ☐ slightly disagree ☐ slightly agree ☐ agree ☐ strongly agree

10. The robo-advisor recommends portfolio D. Are you more or less likely to choose portfolio D?
☐ much less likely ☐ somewhat less likely ☐ slightly less likely ☐ neutral ☐ slightly more likely ☐ somewhat more likely ☐ much more likely

Scenario 2 – Human advisor

Now, suppose that instead of a robo-advisor, this platform provides you with a free human financial advisor, who can analyse your situation and give financial advice.

11. I believe that human financial advisors perform their intended function well.
☐ strongly disagree ☐ disagree ☐ neutral ☐ agree ☐ strongly agree

12. I believe that human financial advisors work in my interest.
☐ strongly disagree ☐ disagree ☐ neutral ☐ agree ☐ strongly agree

Scenario 2.1

Imagine that you are struggling to choose between two portfolios, E or F. Based on your choice, you could gain or lose up to 5% of your initial investment.

13. I would allow the human financial advisor to make the decision on my behalf.
☐ strongly disagree ☐ disagree ☐ slightly disagree ☐ slightly agree ☐ agree ☐ strongly agree

14. The human financial advisor recommends portfolio F. Are you more or less likely to choose portfolio F?
☐ much less likely ☐ somewhat less likely ☐ slightly less likely ☐ neutral ☐ slightly more likely ☐ somewhat more likely ☐ much more likely

Scenario 2.2

Imagine that you are struggling to choose between two portfolios, G or H. Based on your choice, you could gain or lose up to 40% of your initial investment.

15. I would allow the human financial advisor to make the decision on my behalf.
☐ strongly disagree ☐ disagree ☐ slightly disagree ☐ slightly agree ☐ agree ☐ strongly agree

16. The human financial advisor recommends portfolio G. Are you more or less likely to choose portfolio G?
☐ much less likely ☐ somewhat less likely ☐ slightly less likely ☐ neutral ☐ slightly more likely ☐ somewhat more likely ☐ much more likely

Figure A1. Recreation of the Google Forms survey used in this study. Questions are numbered in the order administered; the first question requesting participants' phone numbers for compensation is omitted.

Appendix B

ID	Exp	Subj	AnxL	AnxH	AlCp	AlBn	AIDL	AILL	AIDH	AIH	HuCp	HuBn	HuDL	HuLL	HuDH	HuIH
1	3	2	2	3	4	5	1	4	3	4	5	4	4	6	1	4
2	1	2	3	4	4	4	5	6	3	5	4	4	5	6	5	5
3	1	3	2	4	4	3	4	5	3	3	3	2	3	3	4	3
4	1	3	2	4	3	4	5	6	4	6	4	4	4	5	4	6
5	3	1	2	3	3	4	2	5	2	6	3	4	4	5	2	6
6	2	2	2	3	3	3	4	5	2	4	3	2	4	5	3	4
7	3	1	1	3	3	4	4	5	3	4	4	4	4	5	3	5
8	2	3	2	4	4	2	5	5	2	2	4	2	5	7	3	2
9	1	3	2	4	3	3	4	5	4	1	4	4	5	4	4	3
10	1	4	4	5	2	4	4	5	1	4	5	5	5	7	3	5
11	1	3	2	4	4	3	5	6	1	5	4	4	6	7	5	6
12	1	3	3	5	3	4	6	7	3	5	4	4	5	5	4	6
13	1	3	2	3	3	3	4	5	5	4	4	3	4	6	4	6
14	3	4	1	3	1	1	1	4	1	4	2	2	2	4	2	4
15	3	2	2	3	3	3	4	5	3	3	3	2	5	6	3	3
16	4	2	3	5	3	2	2	4	1	4	4	2	2	5	1	4
17	1	4	2	4	3	2	4	5	2	3	3	2	4	5	2	3
18	2	2	2	4	3	4	4	5	3	5	3	4	5	6	4	5
19	1	2	1	3	1	1	2	4	1	4	3	3	4	4	1	4
20	3	1	1	3	3	4	4	5	2	5	2	2	3	4	2	4
21	3	3	3	3	3	3	4	4	4	4	3	3	4	4	4	4
22	4	2	1	1	3	2	3	4	2	4	2	2	2	4	2	4
23	2	4	2	5	2	3	3	5	1	5	4	3	5	6	1	6
24	4	2	3	3	2	2	2	2	2	2	2	2	2	2	2	2
25	1	2	2	4	3	3	5	5	1	1	4	4	5	7	3	2
26	1	1	5	5	2	1	6	6	1	1	4	4	1	1	1	1
27	2	1	4	4	3	1	1	2	1	2	4	4	5	6	1	5
28	4	2	3	4	4	4	4	5	2	2	2	2	2	4	2	1
29	1	4	1	5	3	3	3	5	1	4	4	3	3	6	1	5
30	2	2	4	5	4	3	4	5	4	4	4	3	4	5	4	5
31	1	5	3	5	2	4	4	7	2	4	5	2	5	7	4	6
32	1	2	2	4	3	3	4	5	4	5	4	4	4	5	4	5
33	3	2	1	3	3	3	4	5	2	4	4	4	4	4	1	5
34	1	2	3	5	4	4	4	5	4	5	4	4	5	6	4	6
35	2	2	1	3	4	4	5	6	5	5	4	4	5	6	5	5
36	1	2	3	5	4	4	4	5	4	5	4	4	5	6	5	6
37	2	2	1	4	3	3	5	5	2	4	4	4	5	6	3	4
38	4	2	1	4	3	3	4	5	1	2	4	2	4	5	1	2
39	5	2	2	5	3	4	2	5	4	7	3	4	5	6	3	5
40	1	1	4	5	4	3	5	6	4	6	3	3	4	5	3	5
41	4	2	2	3	3	3	4	5	3	5	4	3	4	5	4	6
42	4	3	1	2	4	4	4	3	2	3	3	3	4	4	4	4

Scale anchors

Exp (investment experience): 1 = Not at all, 5 = Very much | Subj (perceived subjectivity): 1 = Mostly objective, 5 = Mostly subjective

Anx (anxiety, L = low stakes, H = high stakes): 1 = Not at all, 5 = Extremely anxious

Column abbreviations: Al = AI advisor, Hu = human advisor; Cp = competence/ability, Bn = benevolence;

D = delegation, I = influence; L = low stakes, H = high stakes. Items: 1 = Strongly disagree, 5 = Strongly agree (trust),

1 = Strongly disagree, 6 = Strongly agree (delegation) | Inf = influence: 1 = Much less likely, 7 = Much more likely

Full item wording is given in Appendix A (survey instrument).

Figure B1. Raw survey responses of all 42 participants, recoded to numeric scale values. Column abbreviations and scales are defined in the legend; full item wording is provided in Appendix A.

Appendix C

The project was funded by a fixed budget of \$80 (SGD). The 34 participants who accepted the \$1 financial incentive were compensated via PayLah/PayNow transfer within 12 days from the date the survey was released. The remaining \$46 was redistributed evenly among the five group members for refreshments (\$9.20 each). Table C1 provides a detailed breakdown of the budget allocation.

Item	Cost (SGD)
Participant incentive ($\$1 \times 34$ accepted)	\$34
Total expenditure	\$34
Budget	\$80
Remaining	\$46

Table C1. Project budget and expenditures.

Appendix D

Name	Contribution
Jarren Ng	Topic ideation, literature review, discussion section, proof-reading
Sng Zhenhao	Literature review, proof-reading, initial data analysis, initial draft to introduction
Korakrit Tonpongphan	Topic ideation, Literature review, proof-reading/formatting
Purinut Larpsiriwatkul	Topic ideation, Literature review, proof-reading (survey & report)
Yip Jing Han	Literature review, experiment design, data analysis, diagrams, proof-reading

Table D1. Names and individual contributions of all team members.